Local global hybrid stabilization via a quadratic Lyapunov function and a transport PDE

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Consider the system

$$\dot{x} = f(x) + G(x)u, \qquad x \in \mathbb{R}^n, \quad u \in \mathbb{R}^m, \tag{1}$$

where $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$, $G \in C^1(\mathbb{R}^n, \mathbb{R}^{n\times m})$, and $f(0) = 0$. The objective is to stabilize the equilibrium $x = 0$.

Set

$$A := Df(0), \qquad B := G(0).$$

Assumption 1. The pair (A, B) is stabilizable. That is, there exists a matrix $K \in \mathbb{R}^{m\times n}$ such that

$$A_{\mathrm{cl}} := A + BK$$

is a Hurwitz matrix.

In the inner region, we use the law

$$u_{\mathrm{loc}}(x) = Kx. \tag{2}$$

Lemma 1. *Suppose Assumption 1 holds. Then there exist a symmetric positive definite matrix $P = P^\top > 0$, a number $a > 0$, and a number $c > 0$ such that for the function*

$$W(x) := x^\top P x \tag{3}$$

the estimate

$$L_{f+Gu_{\mathrm{loc}}} W(x) \le -c\, W(x), \qquad 0 < W(x) \le a \tag{4}$$

holds. In particular, the set

$$\Omega_a := \{x \in \mathbb{R}^n : W(x) \le a\}, \qquad \Sigma_a := \{x \in \mathbb{R}^n : W(x) = a\} \tag{5}$$

is positively invariant for the system

$$\dot{x} = f(x) + G(x)u_{\mathrm{loc}}(x),$$

and the equilibrium $x = 0$ is asymptotically stable in Ω_a.

Proof. Since A_{cl} is a Hurwitz matrix, for any $Q = Q^\top > 0$ the Lyapunov equation

$$A_{\mathrm{cl}}^\top P + P A_{\mathrm{cl}} = -Q$$

has a unique solution $P = P^\top > 0$. Fix $Q = I$.

Set

$$F_{\mathrm{loc}}(x) := f(x) + G(x)Kx.$$

Then

$$F_{\mathrm{loc}}(x) = A_{\mathrm{cl}}x + o(\|x\|), \qquad x \to 0.$$

Therefore,

$$\begin{aligned} L_{F_{\mathrm{loc}}}W(x) &= \nabla W(x)^\top F_{\mathrm{loc}}(x) \\ &= 2x^\top P\big(A_{\mathrm{cl}}x + o(\|x\|)\big) \\ &= x^\top (A_{\mathrm{cl}}^\top P + PA_{\mathrm{cl}})x + o(\|x\|^2) \\ &= -\|x\|^2 + o(\|x\|^2). \end{aligned}$$

Hence there exist $r > 0$ and $\mu > 0$ such that for $0 < \|x\| < r$,

$$L_{F_{\mathrm{loc}}}W(x) \le -\mu \|x\|^2 .$$

Since $P > 0$, there exist $\lambda_{\min}, \lambda_{\max} > 0$ such that

$$\lambda_{\min} \|x\|^2 \le W(x) \le \lambda_{\max} \|x\|^2 .$$

Therefore, one can choose $a > 0$ so that $W(x) \le a$ implies $\|x\| < r$, and then

$$L_{F_{\mathrm{loc}}}W(x) \le -\frac{\mu}{\lambda_{\max}} W(x) =: -c\,W(x), \qquad 0 < W(x) \le a.$$

Thus (4) has been obtained.

On the boundary Σ_a, we have

$$L_{F_{\mathrm{loc}}}W(x) \le -ca < 0,$$

so the vector field points inward Ω_a, and Ω_a is positively invariant. Asymptotic stability in Ω_a follows from the direct Lyapunov method. □

The outer region is defined as

$$D_a := \mathbb{R}^n \setminus \Omega_a = \{x \in \mathbb{R}^n : W(x) > a\}.$$

In the outer region, introduce the function

$$T(x) := \log \frac{W(x)}{a}, \qquad x \in D_a. \tag{6}$$

Then

$$T \in C^\infty(D_a), \qquad T > 0 \text{ on } D_a, \qquad T = 0 \text{ on } \Sigma_a,$$

and

$$\nabla T(x) = \frac{\nabla W(x)}{W(x)}. \tag{7}$$

Assumption 2. For all $x \in D_a$, the condition

$$G(x)^\top \nabla W(x) \ne 0 \tag{8}$$

holds.

The outer control is defined by

$$u_{\mathrm{ext}}(x) := -\frac{W(x) + \nabla W(x)^\top f(x)}{\|G(x)^\top \nabla W(x)\|^2}\, G(x)^\top \nabla W(x), \qquad x \in D_a. \tag{9}$$

Lemma 2. *Suppose Assumption 2 holds. Then for all $x \in D_a$,*

$$\nabla T(x)^\top \big(f(x) + G(x)u_{\text{ext}}(x)\big) = -1. \tag{10}$$

Equivalently,

$$L_{f+Gu_{\text{ext}}} W(x) = -W(x), \qquad x \in D_a. \tag{11}$$

Proof. Set

$$b(x) := G(x)^\top \nabla W(x) \in \mathbb{R}^m.$$

Then, by (9),

$$u_{\text{ext}}(x) = -\frac{W(x) + \nabla W(x)^\top f(x)}{\|b(x)\|^2} b(x).$$

Therefore,

$$\begin{aligned}
\nabla W(x)^\top G(x) u_{\text{ext}}(x) &= b(x)^\top u_{\text{ext}}(x) \\
&= -\frac{W(x) + \nabla W(x)^\top f(x)}{\|b(x)\|^2}\, b(x)^\top b(x) \\
&= -W(x) - \nabla W(x)^\top f(x).
\end{aligned}$$

Hence

$$\nabla W(x)^\top \big(f(x) + G(x)u_{\text{ext}}(x)\big) = -W(x),$$

which proves (11).

Using (7), we obtain

$$\nabla T(x)^\top \big(f(x) + G(x)u_{\text{ext}}(x)\big) = \frac{1}{W(x)} \nabla W(x)^\top \big(f(x) + G(x)u_{\text{ext}}(x)\big) = -1.$$

Thus (10) is proved. □

Lemma 3. *Let $x(\cdot)$ be a solution of the outer system*

$$\dot{x} = f(x) + G(x)u_{\text{ext}}(x), \qquad x(0) = x_0 \in D_a.$$

Then, as long as the trajectory remains in D_a,

$$W(x(t)) = W(x_0)e^{-t}. \tag{12}$$

In particular, the trajectory reaches Σ_a in finite time

$$\tau_{\text{in}}(x_0) = \log \frac{W(x_0)}{a}. \tag{13}$$

Proof. By Lemma 2,

$$\frac{d}{dt} W(x(t)) = -W(x(t)).$$

Therefore,

$$W(x(t)) = W(x_0)e^{-t}.$$

The first hitting time of Σ_a is determined by the condition

$$W(x_0)e^{-\tau} = a,$$

hence

$$\tau = \log \frac{W(x_0)}{a}.$$

□

Lemma 4. *For every $x_0 \in D_a$, the outer closed-loop system*

$$\dot{x} = f(x) + G(x)u_{\text{ext}}(x)$$

admits a unique solution on the whole interval

$$[0, \tau_{\text{in}}(x_0)].$$

Moreover, this solution remains in D_a for all $t \in [0, \tau_{\text{in}}(x_0))$ and reaches Σ_a exactly at $t = \tau_{\text{in}}(x_0)$.

Proof. Define

$$F_{\text{ext}}(x) := f(x) + G(x)u_{\text{ext}}(x), \qquad x \in D_a.$$

Since $f \in C^1$, $G \in C^1$, and

$$u_{\text{ext}}(x) = -\frac{W(x) + \nabla W(x)^\top f(x)}{\|G(x)^\top \nabla W(x)\|^2}\, G(x)^\top \nabla W(x),$$

Assumption 2 implies that F_{ext} is C^1 on D_a. In particular, F_{ext} is locally Lipschitz on D_a. Hence for every $x_0 \in D_a$ there exists a unique maximal solution

$$x : [0, t_{\max}) \to D_a$$

of the outer system.

By Lemma 2, along every such solution one has

$$\frac{d}{dt} W(x(t)) = -W(x(t)).$$

Therefore,

$$W(x(t)) = W(x_0)e^{-t}, \qquad t \in [0, t_{\max}).$$

Let

$$\tau_{\text{in}}(x_0) := \log \frac{W(x_0)}{a}.$$

Then for every $t \in [0, \tau_{\text{in}}(x_0))$,

$$W(x(t)) = W(x_0)e^{-t} > a,$$

hence

$$x(t) \in D_a.$$

Thus the trajectory cannot reach Σ_a before $t = \tau_{\text{in}}(x_0)$.

Now fix any $\varepsilon \in (0, \tau_{\text{in}}(x_0))$. For $t \in [0, \tau_{\text{in}}(x_0) - \varepsilon]$,

$$W(x(t)) = W(x_0)e^{-t} \in [ae^{\varepsilon},\ W(x_0)].$$

Hence

$$x([0, \tau_{\text{in}}(x_0) - \varepsilon]) \subset K_\varepsilon(x_0) := \{x \in \mathbb{R}^n :\ ae^{\varepsilon} \le W(x) \le W(x_0)\}.$$

Since $W(x) = x^\top P x$ and $P > 0$, the set $K_\varepsilon(x_0)$ is compact, and moreover

$$K_\varepsilon(x_0) \subset D_a.$$

Therefore F_{ext} is bounded and Lipschitz on a neighborhood of $K_\varepsilon(x_0)$, so the solution can be continued at least up to time $\tau_{\text{in}}(x_0) - \varepsilon$. Since $\varepsilon > 0$ is arbitrary, it follows that

$$t_{\max} \ge \tau_{\text{in}}(x_0).$$

Finally,

$$W(x(\tau_{\text{in}}(x_0))) = W(x_0)e^{-\tau_{\text{in}}(x_0)} = a,$$

hence

$$x(\tau_{\text{in}}(x_0)) \in \Sigma_a.$$

□

The control is applied according to the following rule:

$$u(t) = \begin{cases} u_{\text{ext}}(x(t)), & 0 \le t < \tau_{\text{in}}(x_0), \quad x_0 \in D_a, \\ u_{\text{loc}}(x(t)), & t \ge \tau_{\text{in}}(x_0), \quad x_0 \in D_a, \\ u_{\text{loc}}(x(t)), & t \ge 0, \quad x_0 \in \Omega_a. \end{cases} \tag{14}$$

Theorem 1. *Consider system* (1)*. Assume that:*

1. *Assumption 1 holds;*
2. *$P > 0$, $a > 0$, and $c > 0$ are chosen as in Lemma 1;*
3. *Assumption 2 holds;*
4. *the outer control is given by* (9)*;*
5. *the control is applied according to* (14)*.*

Then the equilibrium $x = 0$ is globally asymptotically stable.

Proof. If $x_0 \in \Omega_a$, then by Lemma 1, the set Ω_a is positively invariant, and the locally closed-loop system is asymptotically stable in Ω_a. Therefore, the solution exists for all $t \ge 0$ and

$$x(t) \to 0 \qquad (t \to +\infty).$$

Now let $x_0 \in D_a$. Then by Lemma 4, the solution of the outer system exists on the interval $[0, \tau_{\text{in}}(x_0)]$, and by Lemma 3,

$$x(\tau_{\text{in}}(x_0)) \in \Sigma_a \subset \Omega_a.$$

After that, the local stabilizer u_{loc} is switched on according to (14). Since Ω_a is positively invariant for the local system, the trajectory remains in Ω_a and converges to zero. Consequently, for every $x_0 \in \mathbb{R}^n$,

$$x(t) \to 0 \qquad (t \to +\infty).$$

It remains to prove Lyapunov stability. Since $P > 0$, there exists $\delta > 0$ such that

$$\|x\| < \delta \quad \Longrightarrow \quad W(x) < a.$$

Therefore, if $\|x_0\| < \delta$, then $x_0 \in \Omega_a$, and the trajectory evolves from the very beginning under the local stabilizer u_{loc}. By local asymptotic stability in Ω_a, the origin is stable.

Thus, the origin is stable and globally attractive, hence it is globally asymptotically stable. □

Remark 1. *The outer law* (9) *can be rewritten in terms of T. Since*

$$\nabla T(x) = \frac{\nabla W(x)}{W(x)},$$

introducing

$$\beta(x) := G(x)^\top \nabla T(x),$$

we obtain the equivalent formula

$$u_{\text{ext}}(x) = -\frac{1 + \nabla T(x)^\top f(x)}{\|\beta(x)\|^2}\, \beta(x).$$

Then automatically

$$\nabla T(x)^\top (f(x) + G(x) u_{\text{ext}}(x)) = -1.$$

For the choice $T(x) = \log \frac{W(x)}{a}$, this is equivalent to

$$\dot{W} = -W$$

in the outer region.

1 Example: two-tank hydraulic system

Consider a system of two interconnected tanks. Let

$$x = \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} \in \mathbb{R}^2$$

denote the deviations of the liquid levels from a prescribed operating point, and let

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \in \mathbb{R}^2$$

denote the control inflows into each tank.

The system dynamics are

$$\dot{x} = f(x) + u, \tag{15}$$

where

$$f(x) = \begin{bmatrix} \alpha \dfrac{h_1^2}{r^2 + h_1^2}\, h_1 + \kappa(h_2 - h_1) \\ \alpha \dfrac{h_2^2}{r^2 + h_2^2}\, h_2 + \kappa(h_1 - h_2) \end{bmatrix}, \qquad \alpha > 0,\ \kappa > 0,\ r > 0.$$

The control objective is to stabilize the equilibrium

$$x^\star = 0.$$

Local stabilizer

Since

$$\alpha \frac{h_i^2}{r^2 + h_i^2} h_i = O(h_i^3), \qquad h_i \to 0,$$

the linearization of (15) at the origin is

$$A = \begin{bmatrix} -\kappa & \kappa \\ \kappa & -\kappa \end{bmatrix}, \qquad B = I_2.$$

Choose the local linear feedback

$$u_{\text{loc}}(x) = Kx, \qquad K = -kI_2, \quad k > 0.$$

Then

$$A_{\text{cl}} = A + BK = \begin{bmatrix} -(\kappa + k) & \kappa \\ \kappa & -(\kappa + k) \end{bmatrix}.$$

Its eigenvalues are

$$\lambda_1 = -k, \qquad \lambda_2 = -(2\kappa + k),$$

hence A_{cl} is Hurwitz for every $k > 0$.

Therefore there exists a unique matrix $P = P^\top > 0$ solving

$$A_{\text{cl}}^\top P + P A_{\text{cl}} = -I.$$

Since A_{cl} is symmetric, we may seek P in the form

$$P = \begin{bmatrix} p & q \\ q & p \end{bmatrix}.$$

A direct computation yields

$$p = \frac{\kappa + k}{2k(2\kappa + k)}, \qquad q = \frac{\kappa}{2k(2\kappa + k)}.$$

Hence

$$P = \frac{1}{2k(2\kappa + k)} \begin{bmatrix} \kappa + k & \kappa \\ \kappa & \kappa + k \end{bmatrix}. \tag{16}$$

We now define the quadratic Lyapunov function exactly as in the general theorem:

$$W(x) = x^\top P x. \tag{17}$$

By Lemma 1, there exists $a > 0$ such that

$$L_{f+u_{\text{loc}}} W(x) < 0, \qquad 0 < W(x) \le a.$$

Therefore the set

$$\Omega_a = \{x \in \mathbb{R}^2 : \ x^\top P x \le a\}$$

is positively invariant for the local closedloop system, and the origin is asymptotically stable in Ω_a.

Lack of global stability of the local law

The local stabilizer is not necessarily global. Indeed, on the invariant subset $h_1 = h_2 = s$, the system reduces to

$$\dot{s} = \left(\alpha \frac{s^2}{r^2 + s^2} - k \right) s.$$

For sufficiently large $|s|$, one has

$$\dot{s} \approx (\alpha - k)s.$$

Hence if $\alpha > k$, trajectories on this invariant set move away from the origin for large $|s|$, and the local feedback is not globally stabilizing.

Outer control

In the outer region

$$D_a = \{x \in \mathbb{R}^2 : \ x^\top P x > a\},$$

define

$$T(x) = \log \frac{x^\top P x}{a}.$$

Since $G(x) = I_2$, Assumption 2 becomes

$$\nabla W(x) \neq 0 \qquad \text{for all } x \in D_a.$$

This holds because

$$\nabla W(x) = 2Px,$$

and $P > 0$ implies $Px \neq 0$ whenever $x \neq 0$. In particular, $\nabla W(x) \neq 0$ on D_a.

Therefore the outer control is given by

$$u_{\text{ext}}(x) = -\frac{W(x) + \nabla W(x)^\top f(x)}{\|\nabla W(x)\|^2} \nabla W(x), \qquad x \in D_a. \tag{18}$$

Using $\nabla W(x) = 2Px$, this can be rewritten as

$$u_{\text{ext}}(x) = -\frac{x^\top Px + 2x^\top Pf(x)}{2x^\top P^2 x}\, Px, \qquad x \in D_a.$$

By Lemma 2,

$$L_{f+u_{\text{ext}}} W(x) = -W(x), \qquad x \in D_a,$$

and thus along every outer trajectory

$$W(x(t)) = W(x_0)e^{-t}.$$

Hence every trajectory starting from D_a reaches the switching surface

$$\Sigma_a = \{x \in \mathbb{R}^2 : \; x^\top Px = a\}$$

in finite time

$$\tau_{\text{in}}(x_0) = \log \frac{W(x_0)}{a}.$$

Hybrid control law

The overall control is

$$u(t) = \begin{cases} u_{\text{ext}}(x(t)), & 0 \le t < \tau_{\text{in}}(x_0), \quad x_0 \in D_a, \\ u_{\text{loc}}(x(t)), & t \ge \tau_{\text{in}}(x_0), \quad x_0 \in D_a, \\ u_{\text{loc}}(x(t)), & t \ge 0, \quad x_0 \in \Omega_a. \end{cases}$$

Thus every trajectory starting outside Ω_a reaches Σ_a in finite time under the outer law, enters Ω_a, and then converges to the origin under the local stabilizer. By Theorem 1, the equilibrium $x = 0$ is globally asymptotically stable for this hybrid law.

Outer control with tangential swirling component

The outer control defined in (18) is the simplest choice enforcing

$$\nabla T(x)^\top \big(f(x) + u_{\text{ext}}(x)\big) = -1.$$

However, in dimension two one can add a tangential component without changing this identity.

Let

$$g(x) := \nabla T(x) = \frac{\nabla W(x)}{W(x)}.$$

Define the rotated vector

$$g^\perp(x) := \begin{bmatrix} -g_2(x) \\ g_1(x) \end{bmatrix}.$$

Then

$$g(x)^\top g^\perp(x) = 0.$$

Therefore, for any scalar function $\omega : D_a \to \mathbb{R}$, the control

$$u_{\text{ext}}^{\text{sw}}(x) := u_{\text{ext}}(x) + \omega(x)\, g^\perp(x) \tag{19}$$

still satisfies the same transport PDE:

$$\nabla T(x)^\top \big(f(x) + u_{\text{ext}}^{\text{sw}}(x)\big) = -1.$$

Indeed,

$$\begin{aligned}\nabla T(x)^\top \big(f(x) + u_{\text{ext}}^{\text{sw}}(x)\big) &= \nabla T(x)^\top \big(f(x) + u_{\text{ext}}(x)\big) + \omega(x)\, \nabla T(x)^\top g^\perp(x) \\ &= -1 + \omega(x)\, g(x)^\top g^\perp(x) \\ &= -1.\end{aligned}$$

Thus the tangential term does not affect the radial decay of W. In particular,

$$L_{f+u_{\text{ext}}^{\text{sw}}} W(x) = -W(x), \qquad x \in D_a,$$

and therefore

$$W(x(t)) = W(x_0)e^{-t}$$

still holds for every outer trajectory.

A convenient choice is to let the swirl vanish near the switching surface and remain stronger farther away. For example, one may take

$$\omega(x) = \beta\, \frac{W(x) - a}{W(x) + \varepsilon}, \qquad \beta > 0,\ \varepsilon > 0. \tag{20}$$

Then $\omega(x) \to 0$ as $W(x) \downarrow a$, while for larger values of $W(x)$ the tangential motion becomes more pronounced.

Consequently, the outer control with swirl can be written as

$$u_{\text{ext}}^{\text{sw}}(x) = -\frac{W(x) + \nabla W(x)^\top f(x)}{\|\nabla W(x)\|^2}\, \nabla W(x) + \omega(x) \begin{bmatrix} -\partial_{h_2} T(x) \\ \partial_{h_1} T(x) \end{bmatrix}, \qquad x \in D_a. \tag{21}$$

Since $T(x) = \log \frac{W(x)}{a}$, this can be rewritten as

$$u_{\text{ext}}^{\text{sw}}(x) = -\frac{W(x) + \nabla W(x)^\top f(x)}{\|\nabla W(x)\|^2}\, \nabla W(x) + \frac{\omega(x)}{W(x)} \begin{bmatrix} -\partial_{h_2} W(x) \\ \partial_{h_1} W(x) \end{bmatrix}.$$

Hence the swirl term modifies only the motion along the level sets of T and W, while preserving the finite time entrance property into Σ_a.